A SIX-POINT POSITIVE-WEIGHT OBSTRUCTION TO DEGREE-FOUR HERMITE CURVATURE REPAIRS ON MANIFOLDS WITHOUT CONJUGATE POINTS

ARGUS OPEN RESEARCH PROJECT

Abstract. In the construction of Riemannian metrics without conjugate points whose geodesic rays fail opposite-end divergence, we prove that the degree-four compactly supported Hermite transverse blister repair over $[-16, 16]^5$ cannot satisfy the preregistered sectional-curvature gate $K \leq 1$: six positive rational plateau weights simultaneously annihilate all five free coefficient directions and yield a strictly negative weighted slack, forcing every real coefficient vector in $\mathbb{R}^5$ to violate $K \leq 1$ at one of six exact rational evaluation points.

1. Main Farkas Obstruction

Theorem 1.1. *Let g_c ($c \in \mathbb{R}^5$) be the five-parameter degree-four Hermite compactly supported metric perturbation preserving the full first jet along the central geodesic. There exist six explicit rational evaluation points $q_1, \ldots, q_6$ and positive rational weights $w_1, \ldots, w_6 \in \mathbb{Q}_{>0}$ such that the linearized coefficient matrix $B \in \mathbb{Q}^{6\times 5}$ satisfies $w^{\mathsf{T}} B = 0$ and the baseline curvature excess vector b_0 satisfies $w^{\mathsf{T}} b_0 > 0$. Consequently, for every $c \in \mathbb{R}^5$, $\max_{1\leq i\leq 6} K_{g_c}(q_i) > 1$.*

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